
Bayesian Statistics III: Bayesian Linear Regression

ECON 8310: Business Forecasting — Lecture 12

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Lecture Outline

- 1 Bayesian Linear Regression
- 2 Reading the Posterior
- 3 Causal Structure and DAGs
- 4 Course Synthesis and Next Steps

Bayesian Linear Regression

The same regression you already know — returning a distribution for every coefficient.

OLS vs. Bayesian Regression: What Changes?

You have fitted linear models since Lecture 2, and regularized them in Lecture 6. OLS solves for a single best coefficient vector:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

One number per predictor, with a standard error resting on large-sample theory. The Bayesian version replaces that with the update rule from Lecture 10:

$$p(\beta \mid \mathbf{y}) \propto p(\mathbf{y} \mid \beta) \cdot p(\beta)$$

Every coefficient comes back as a distribution. Not an estimate plus an error bar computed from an asymptotic formula — a full posterior you can ask questions of directly.

And one thing you know reappears: **regularization is a prior**. A Normal prior on β is Ridge; a Laplace prior is LASSO. Lecture 6's penalty is, here, a statement of belief.

Log units on log price, SNAP and events

```
with pm.Model() as model:
    a      = pm.Normal("a", y.mean(), 1)
    b_price = pm.Normal("b_price", 0, 1)      # weakly informative
    b_snap  = pm.Normal("b_snap", 0, 1)
    b_event = pm.Normal("b_event", 0, 1)
    sigma   = pm.Exponential("sigma", 1.0)

    mu = a + b_price*lp + b_snap*snap + b_event*event

    pm.Normal("obs", mu=mu, sigma=sigma, observed=y)

    idata = pm.sample(1000, tune=1500, chains=4)
```

Standardize the predictors first, so `Normal(0, 1)` means the same thing for each.

With units and price both in logs, `b_price` is an **elasticity** — the percent change in demand per percent change in price. A quantity a category manager already uses.

Reading the Posterior

Three questions you can ask a distribution that you cannot ask a point estimate.

Reading a Posterior Coefficient

Fitted on the weekly panel, controlling for series and trend:

Coefficient	Mean	94% HDI	Reading
b_price	-0.127	[-0.150, -0.100]	Inelastic, and clearly negative
b_trend	0.169	[0.164, 0.173]	Demand grows over the sample
b_snap	0.026	[0.023, 0.030]	About a 2.6% SNAP-week lift
b_event	-0.002	[-0.005, 0.002]	Spans zero

The **HDI** — highest density interval — is the shortest interval containing 94% of the posterior. Unlike a confidence interval it says what everyone wants it to say: there is a 94% probability the coefficient lies in there.

The last row is the useful one. b_event's interval spans zero, so the data do not establish a direction. That is a more honest report than “not significant,” because it also shows the effect is *small* — at most about half a percent either way.

Asking the Posterior a Direct Question

The posterior is a set of samples, so any question about β is answered by counting.

One line each

```
b = idata.posterior["b_price"].values.ravel()

(b < 0).mean()          # P(demand falls when price rises)
(b < -0.5).mean()       # P(elasticity beyond -0.5)
((b > -0.2) & (b < 0)).mean() # P(inelastic but negative)
```

On our fit, $P(\text{elasticity} < 0) = 1.00$ — every posterior sample is negative, so the direction is not in doubt. But $P(\text{elasticity} < -0.5) \approx 0$: the effect is real and it is *small*.

Two different findings, and a p-value conflates them. “Statistically significant” reports only the first. The business needs the second.

*Asking how much posterior mass falls in a range of negligible values is called a **ROPE** — region of practical equivalence.*

Causal Structure and DAGs

Which coefficients mean what you think they mean.

Directed Acyclic Graphs

A **DAG** is a picture of what you believe causes what: **nodes** are variables, **arrows** run from cause to effect, and **acyclic** means nothing causes itself.

It is drawn before the model, from domain knowledge, and its purpose is to tell you *which variables belong in the regression*.

Confounder	Causes both your predictor and your outcome. Must be controlled for, or its influence is absorbed into your coefficient.
Collider	Is caused by both. Controlling for it creates a spurious association that was not there.

That second row is why “add every control you have” is bad advice. Which variables to include is a question about causal structure, and no amount of model fit will answer it.

Whether a given DAG is right is a substantive question about your business, not a statistical one. This course uses DAGs to reason about what a coefficient means — estimating causal effects credibly is a separate subject.

Confounding, Measured

Our panel has thirty store-category series. Categories differ in *both* typical price and typical volume — so series identity is an arrow into price *and* into units. A textbook confounder.

Fit the same elasticity twice:

Model	Elasticity	94% HDI
Price, SNAP, events only	-0.687	[-0.700, -0.676]
+ series effects and trend	-0.127	[-0.150, -0.100]

The uncontrolled estimate is more than five times too large. It is not noisier — its interval is actually *narrower*. It is confidently, precisely wrong.

A tight credible interval says the model is sure. It says nothing about whether the model is answering your question. Bayesian machinery does not protect you from omitting a confounder — only the DAG does.

Scenario Analysis with Posterior Samples

Because the posterior is samples, you can push a business scenario through it and get a **distribution of outcomes** rather than a single projected number.

What if we raise price 5%?

```
b = idata.posterior["b_price"].values.ravel()
pct_change = 100 * (np.exp(b * np.log(1.05)) - 1)
pct_change.mean(), np.percentile(pct_change, [3, 97])
```

With an elasticity near -0.127 , a 5% price rise implies roughly a **0.6% fall in units** — and returns the whole distribution of that fall, so you can report the pessimistic end too.

Then the decision follows. Revenue rises if the volume loss is smaller than the price gain, and you can compute that probability directly.

Valid only to the extent the DAG is — this is a causal claim about intervening on price.

Course Synthesis and Next Steps

From classical time series to Bayesian hierarchical models: a complete forecasting toolkit.

Course Method Map

Method	Data	Uncertainty	Interpretable	Strengths
ETS / ARIMA (L01–02)	Univariate TS	CI	High	Fast, automatic
GAM / Prophet (L03)	TS + predictors	Limited	High	Nonlinear + interpretable
Decision Tree (L04)	Tabular	No	High	Rule-based explanation
Random Forest (L05)	Tabular	OOB CI	Medium	Robust, minimal tuning
XGBoost (L06)	Tabular	No	Low	Best tabular accuracy
FFN / 1D CNN (L07–08)	Tabular / TS	No	Low	Smooth functions, fast
LSTM / Transformer (L09)	Sequences	No	Very low	Long-range dependencies
Bayesian (L10–12)	Any	Full posterior	High	Uncertainty, small samples

No single method dominates. Choose based on: (1) data type and size, (2) need for uncertainty quantification, (3) need for interpretability, (4) time to deploy. In practice, combine: use XGBoost for the point forecast and a Bayesian model for uncertainty intervals.

Lecture 12: Key Takeaways

1. **Bayesian linear regression** places priors on coefficients and returns a full posterior distribution for each β_j . A Normal prior is equivalent to Ridge regularization; a Laplace prior is equivalent to LASSO.
2. Posterior coefficient plots give: posterior mean, HDI, and $P(\beta_j > 0)$ — richer information than a single p -value.
3. **DAGs** make the causal structure explicit, helping identify confounders (must include) and colliders (must exclude). Always draw the DAG before specifying a regression model.
4. **Scenario analysis** with posterior samples: plug any counterfactual $\mathbf{x}^*$ into posterior draws to get a full predictive distribution for that scenario.
5. **No single method is best for all forecasting problems.** Build intuition for when to use each tool in this course's toolkit, and combine methods (e.g., point forecast + Bayesian uncertainty) when the decision stakes are high.

References I
